

Terraform AWS Two-Tier Architecture

Project 1 - Plan, Apply, Live Web Server & Destroy

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Environment	dev (Single-env)
Cloud / Region	AWS - ap-south-1 (Mumbai)
Tool	Terraform, run from Git Bash on Windows
Architecture	VPC 10.0.0.0/16 public + private subnet web + db t3.micro
Resources managed	10 (VPC, IGW, 2 subnets, route table + assoc., 2 security groups, 2 EC2)
Live endpoint	http://13.206.71.186 (web server responded)

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terraform plan

Preview of the 10 resources Terraform will create

pavan@LAPTOP-E8090IMQ MINGW64 ~/Project-1-web-app/Single-env (master)

\$ terraform plan

data.aws_availability_zones.available: Reading...

data.aws_ami.amazon_linux: Reading...

data.aws_availability_zones.available: Read complete after 0s [id=ap-south-1]

data.aws_ami.amazon_linux: Read complete after 0s [id=ami-03054015e26069645]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

aws_instance.db will be created

```
+ resource "aws_instance" "db" {
  + ami                  = "ami-03054015e26069645"
  + arn                  = (known after apply)
  + associate_public_ip_address = false
  + availability_zone     = (known after apply)
  + cpu_core_count        = (known after apply)
  + cpu_threads_per_core   = (known after apply)
  + disable_api_stop       = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized          = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + get_password_data       = false
  + host_id                = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                     = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
  + ipv6_address_count       = (known after apply)
  + ipv6_addresses          = (known after apply)
  + key_name                = (known after apply)
  + monitoring              = (known after apply)
  + outpost_arn             = (known after apply)
  + password_data           = (known after apply)
  + placement_group         = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns             = (known after apply)
  + private_ip              = (known after apply)
  + public_dns              = (known after apply)
```

```
+ public_ip                = (known after apply)
+ secondary_private_ips    = (known after apply)
+ security_groups          = (known after apply)
+ source_dest_check        = true
+ spot_instance_request_id = (known after apply)
+ subnet_id               = (known after apply)
+ tags                    = {
  + "Environment" = "dev"
  + "Name"        = "pavan-two-tier-db-server"
  + "Owner"       = "Pavan"
}
+ tags_all                 = {
  + "Environment" = "dev"
  + "Name"        = "pavan-two-tier-db-server"
  + "Owner"       = "Pavan"
}
+ tenancy                  = (known after apply)
+ user_data                = (known after apply)
+ user_data_base64        = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids   = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}
```

aws_instance.web will be created

```

+ resource "aws_instance" "web" {
  + ami                  = "ami-03054015e26069645"
  + arn                  = (known after apply)
  + associate_public_ip_address = true
  + availability_zone     = (known after apply)
  + cpu_core_count        = (known after apply)
  + cpu_threads_per_core  = (known after apply)
  + disable_api_stop      = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized         = (known after apply)
  + enable_primary_ipv6   = (known after apply)
  + get_password_data     = false
  + host_id               = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile   = (known after apply)
  + id                    = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle    = (known after apply)
  + instance_state        = (known after apply)
  + instance_type         = "t3.micro"
  + ipv6_address_count     = (known after apply)
  + ipv6_addresses        = (known after apply)
  + key_name               = (known after apply)
  + monitoring             = (known after apply)
  + outpost_arn            = (known after apply)
  + password_data          = (known after apply)
  + placement_group        = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns            = (known after apply)
  + private_ip             = (known after apply)
  + public_dns             = (known after apply)
  + public_ip              = (known after apply)
  + secondary_private_ips  = (known after apply)
  + security_groups        = (known after apply)
  + source_dest_check      = true
  + spot_instance_request_id = (known after apply)
  + subnet_id              = (known after apply)
  + tags                   = {
    + "Environment" = "dev"
    + "Name"        = "pavan-two-tier-web-server"
    + "Owner"       = "Pavan"
  }
+ tags_all                = {
  + "Environment" = "dev"
  + "Name"        = "pavan-two-tier-web-server"

```

```

    + "Owner"      = "Pavan"
  }
+ tenancy          = (known after apply)
+ user_data        = "2a6dc73cb3ff7fe2af9f6bee13a158d650f8000"
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = true
+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}

# aws_internet_gateway.main will be created
+ resource "aws_internet_gateway" "main" {
+   arn      = (known after apply)
+   id       = (known after apply)
+   owner_id = (known after apply)
+   tags     = {
+     + "Environment" = "dev"
+     + "Name"        = "pavan-two-tier-igw"
+     + "Owner"       = "Pavan"
+   }
+   tags_all = {
+     + "Environment" = "dev"
+     + "Name"        = "pavan-two-tier-igw"
+     + "Owner"       = "Pavan"
+   }
}
```

```

    + vpc_id    = (known after apply)
  }

# aws_route_table.public will be created
+ resource "aws_route_table" "public" {
  + arn                = (known after apply)
  + id                 = (known after apply)
  + owner_id           = (known after apply)
  + propagating_vgws   = (known after apply)
  + route              = [
    + {
      + cidr_block      = "0.0.0.0/0"
      + gateway_id      = (known after apply)
      # (11 unchanged attributes hidden)
    },
  ]
  + tags               = {
    + "Environment" = "dev"
    + "Name"        = "pavan-two-tier-public-rt"
    + "Owner"       = "Pavan"
  }
  + tags_all           = {
    + "Environment" = "dev"
    + "Name"        = "pavan-two-tier-public-rt"
    + "Owner"       = "Pavan"
  }
  + vpc_id             = (known after apply)
}

# aws_route_table_association.public will be created
+ resource "aws_route_table_association" "public" {
  + id                 = (known after apply)
  + route_table_id     = (known after apply)
  + subnet_id          = (known after apply)
}

# aws_security_group.db will be created
+ resource "aws_security_group" "db" {
  + arn                = (known after apply)
  + description         = "Allow MySQL only from the web security group"
  + egress              = [
    + {
      + cidr_blocks     = [
        + "0.0.0.0/0",
      ]
      + description     = "Allow outbound traffic"
    }
  ]
}

```

```

    + from_port      = 0
    + ipv6_cidr_blocks = []
    + prefix_list_ids = []
    + protocol       = "-1"
    + security_groups = []
    + self           = false
    + to_port        = 0
  },
]
+ id = (known after apply)
+ ingress = [
  + {
    + cidr_blocks      = []
    + description      = "MySQL from web server security group"
    + from_port        = 3306
    + ipv6_cidr_blocks = []
    + prefix_list_ids  = []
    + protocol         = "tcp"
    + security_groups  = (known after apply)
    + self             = false
    + to_port          = 3306
  },
]
+ name = "pavan-two-tier-db-sg"
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ revoke_rules_on_delete = false
+ tags = {
  + "Environment" = "dev"
  + "Name"        = "pavan-two-tier-db-sg"
  + "Owner"       = "Pavan"
}
+ tags_all = {
  + "Environment" = "dev"
  + "Name"        = "pavan-two-tier-db-sg"
  + "Owner"       = "Pavan"
}
+ vpc_id = (known after apply)
}

```

aws_security_group.web will be created

```

+ resource "aws_security_group" "web" {
  + arn = (known after apply)
  + description = "Allow HTTP and restricted SSH access"
  + egress = [
    + {

```

```

    + cidr_blocks      = [
      + "0.0.0.0/0",
    ]
    + description      = "Allow outbound traffic"
    + from_port        = 0
    + ipv6_cidr_blocks = []
    + prefix_list_ids  = []
    + protocol         = "-1"
    + security_groups  = []
    + self             = false
    + to_port          = 0
  },
]
+ id                        = (known after apply)
+ ingress                  = [
  + {
    + cidr_blocks      = [
      + "0.0.0.0/0",
    ]
    + description      = "HTTP from the internet"
    + from_port        = 80
    + ipv6_cidr_blocks = []
    + prefix_list_ids  = []
    + protocol         = "tcp"
    + security_groups  = []
    + self             = false
    + to_port          = 80
  },
  + {
    + cidr_blocks      = [
      + "106.222.228.188/32",
    ]
    + description      = "SSH from my IP only"
    + from_port        = 22
    + ipv6_cidr_blocks = []
    + prefix_list_ids  = []
    + protocol         = "tcp"
    + security_groups  = []
    + self             = false
    + to_port          = 22
  },
]
+ name                     = "pavan-two-tier-web-sg"
+ name_prefix              = (known after apply)
+ owner_id                 = (known after apply)
+ revoke_rules_on_delete  = false

```

```

+ tags              = {
+   "Environment" = "dev"
+   "Name"        = "pavan-two-tier-web-sg"
+   "Owner"       = "Pavan"
+ }
+ tags_all          = {
+   "Environment" = "dev"
+   "Name"        = "pavan-two-tier-web-sg"
+   "Owner"       = "Pavan"
+ }
+ vpc_id            = (known after apply)
}

# aws_subnet.private will be created
+ resource "aws_subnet" "private" {
+   arn                                = (known after apply)
+   assign_ipv6_address_on_creation    = false
+   availability_zone                 = "ap-south-1a"
+   availability_zone_id               = (known after apply)
+   cidr_block                        = "10.0.2.0/24"
+   enable_dns64                      = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id                                = (known after apply)
+   ipv6_cidr_block_association_id     = (known after apply)
+   ipv6_native                       = false
+   map_public_ip_on_launch           = false
+   owner_id                          = (known after apply)
+   private_dns_hostname_type_on_launch = (known after apply)
+   tags                              = {
+     "Environment" = "dev"
+     "Name"        = "pavan-two-tier-private-subnet"
+     "Owner"       = "Pavan"
+   }
+   tags_all                        = {
+     "Environment" = "dev"
+     "Name"        = "pavan-two-tier-private-subnet"
+     "Owner"       = "Pavan"
+   }
+   vpc_id                        = (known after apply)
+ }

# aws_subnet.public will be created
+ resource "aws_subnet" "public" {
+   arn                                = (known after apply)
+   assign_ipv6_address_on_creation    = false

```

```

+ availability_zone           = "ap-south-1a"
+ availability_zone_id        = (known after apply)
+ cidr_block                  = "10.0.1.0/24"
+ enable_dns64                = false
+ enable_resource_name_dns_a_record_on_launch = false
+ enable_resource_name_dns_aaaa_record_on_launch = false
+ id                          = (known after apply)
+ ipv6_cidr_block_association_id = (known after apply)
+ ipv6_native                 = false
+ map_public_ip_on_launch     = true
+ owner_id                   = (known after apply)
+ private_dns_hostname_type_on_launch = (known after apply)
+ tags                        = {
  + "Environment" = "dev"
  + "Name"         = "pavan-two-tier-public-subnet"
  + "Owner"        = "Pavan"
}
+ tags_all                    = {
  + "Environment" = "dev"
  + "Name"         = "pavan-two-tier-public-subnet"
  + "Owner"        = "Pavan"
}
+ vpc_id                      = (known after apply)
}

```

aws_vpc.main will be created

```

+ resource "aws_vpc" "main" {
  + arn                = (known after apply)
  + cidr_block         = "10.0.0.0/16"
  + default_network_acl_id = (known after apply)
  + default_route_table_id = (known after apply)
  + default_security_group_id = (known after apply)
  + dhcp_options_id     = (known after apply)
  + enable_dns_hostnames = true
  + enable_dns_support   = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                  = (known after apply)
  + instance_tenancy     = "default"
  + ipv6_association_id  = (known after apply)
  + ipv6_cidr_block      = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id  = (known after apply)
  + owner_id             = (known after apply)
  + tags                = {
    + "Environment" = "dev"
    + "Name"         = "pavan-two-tier-vpc"
  }
}

```

```
    + "Owner"      = "Pavan"
  }
+ tags_all          = {
  + "Environment" = "dev"
  + "Name"        = "pavan-two-tier-vpc"
  + "Owner"       = "Pavan"
}
}
```

Plan: 10 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ db_private_ip = (known after apply)
+ vpc_id        = (known after apply)
+ web_public_ip = (known after apply)
+ web_public_url = (known after apply)
```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Releasing state lock. This may take a few moments...

2

terraform apply

10 resources created in ~25 seconds; outputs returned

pavan@LAPTOP-E8090IMQ MINGW64 ~/Project-1-web-app/Single-env (master)

\$ terraform apply

data.aws_availability_zones.available: Reading...

data.aws_ami.amazon_linux: Reading...

data.aws_availability_zones.available: Read complete after 0s [id=ap-south-1]

data.aws_ami.amazon_linux: Read complete after 1s [id=ami-03054015e26069645]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

[... detailed resource plan identical to Section 1 (terraform plan) omitted here to keep this page set short ...]

Plan: 10 to add, 0 to change, 0 to destroy.

Changes to Outputs:

+ db_private_ip = (known after apply)

+ vpc_id = (known after apply)

+ web_public_ip = (known after apply)

+ web_public_url = (known after apply)

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.main: Creating...

aws_vpc.main: Still creating... [00m10s elapsed]

aws_vpc.main: Creation complete after 12s [id=vpc-08b0e12ad40849d8e]

aws_internet_gateway.main: Creating...

aws_subnet.private: Creating...

aws_subnet.public: Creating...

aws_security_group.web: Creating...

aws_internet_gateway.main: Creation complete after 0s [id=igw-00b8f91164beec053]

aws_route_table.public: Creating...

aws_subnet.private: Creation complete after 1s [id=subnet-0e7bcf7ede6e6e84c]

aws_route_table.public: Creation complete after 1s [id=rtb-074b990bfba86b651]

aws_security_group.web: Creation complete after 2s [id=sg-0280ac5e0c547834d]

aws_security_group.db: Creating...

aws_security_group.db: Creation complete after 2s [id=sg-09cd2d47c5452d6f9]

aws_instance.db: Creating...

aws_subnet.public: Still creating... [00m10s elapsed]

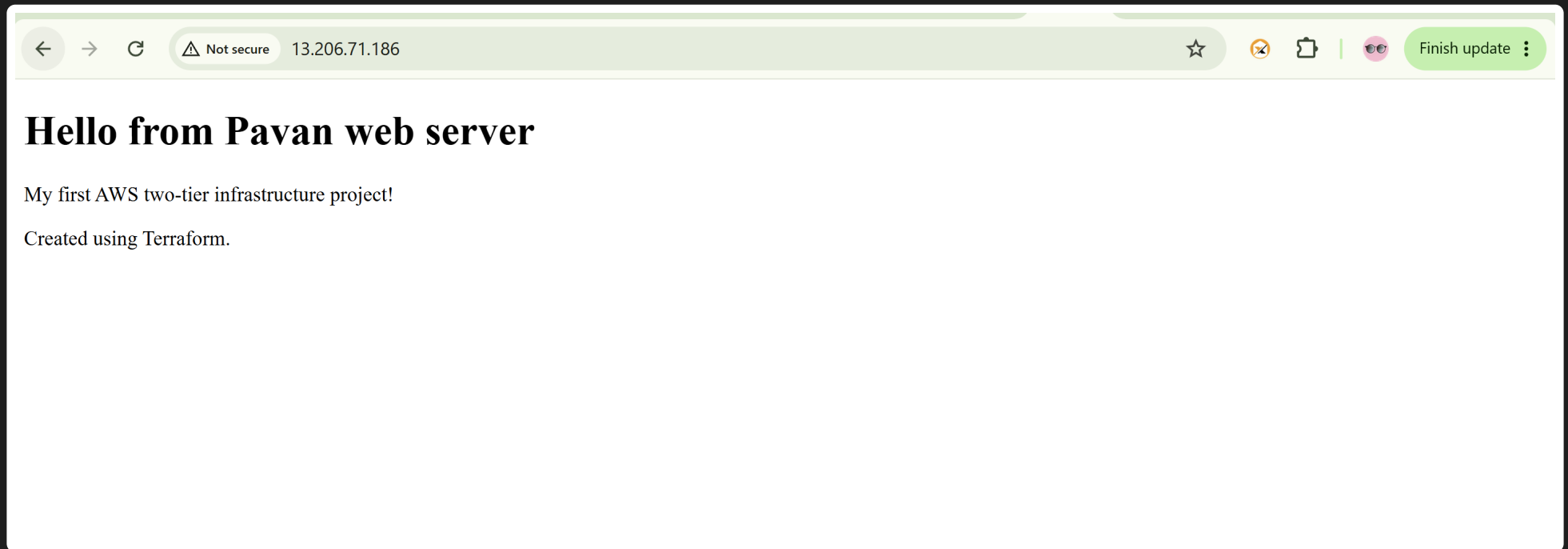
aws_subnet.public: Creation complete after 11s [id=subnet-04b99964e06fc9fb1]

```
aws_route_table_association.public: Creating...
aws_instance.web: Creating...
aws_route_table_association.public: Creation complete after 0s [id=rtbassoc-0091e4d387ad8d9d6]
aws_instance.db: Still creating... [00m10s elapsed]
aws_instance.db: Creation complete after 13s [id=i-08f53c320d3a49ce1]
aws_instance.web: Still creating... [00m10s elapsed]
aws_instance.web: Creation complete after 12s [id=i-08c0c7d3627c6a2c9]
Releasing state lock. This may take a few moments...
```

Apply complete! Resources: 10 added, 0 changed, 0 destroyed.

Outputs:

```
db_private_ip = "10.0.2.102"
vpc_id = "vpc-08b0e12ad40849d8e"
web_public_ip = "13.206.71.186"
web_public_url = "http://13.206.71.186"
```



Result: the EC2 web server deployed by Terraform is reachable from the internet

URL opened: `http://13.206.71.186` (output `web_public_url` from terraform apply)

Page shown: "Hello from Pavan web server - My first AWS two-tier infrastructure project! - Created using Terraform."

Security group allows HTTP (port 80) from the internet; SSH (port 22) only from the owner's IP.

4

terraform destroy

Clean teardown of all 10 resources, then state lock released

pavan@LAPTOP-E8090IMQ MINGW64 ~/Project-1-web-app/Single-env (master)

\$ terraform destroy

```
data.aws_availability_zones.available: Reading...
data.aws_ami.amazon_linux: Reading...
aws_vpc.main: Refreshing state... [id=vpc-0fc2b2a37e1ef0efc]
data.aws_availability_zones.available: Read complete after 0s [id=ap-south-1]
data.aws_ami.amazon_linux: Read complete after 1s [id=ami-03054015e26069645]
aws_internet_gateway.main: Refreshing state... [id=igw-064471428d88acd6a]
aws_subnet.public: Refreshing state... [id=subnet-0e012946894175486]
aws_subnet.private: Refreshing state... [id=subnet-03695136206a6d1c0]
aws_security_group.web: Refreshing state... [id=sg-0cfff1c26ee7cbdfb9]
aws_route_table.public: Refreshing state... [id=rtb-0d4800a7500ab5398]
aws_route_table_association.public: Refreshing state... [id=rtbassoc-08ec72c09adc1b2f4]
aws_security_group.db: Refreshing state... [id=sg-0ef96a959e337c451]
aws_instance.web: Refreshing state... [id=i-009a7068ece7510db]
aws_instance.db: Refreshing state... [id=i-02e89085cd19517c6]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

aws_instance.db will be destroyed

```
- resource "aws_instance" "db" {
  - ami                               = "ami-03054015e26069645" -> null
  - arn                               = "arn:aws:ec2:ap-south-1:987078319571:instance/i-02e89085cd19517c6" -> null
  - associate_public_ip_address      = false -> null
  - availability_zone                 = "ap-south-1a" -> null
  - cpu_core_count                    = 1 -> null
  - cpu_threads_per_core              = 2 -> null
  - disable_api_stop                  = false -> null
  - disable_api_termination           = false -> null
  - ebs_optimized                     = false -> null
  - get_password_data                 = false -> null
  - hibernation                       = false -> null
  - id                               = "i-02e89085cd19517c6" -> null
  - instance_initiated_shutdown_behavior = "stop" -> null
  - instance_state                    = "running" -> null
  - instance_type                     = "t3.micro" -> null
  - ipv6_address_count                 = 0 -> null
  - ipv6_addresses                    = [] -> null
  - monitoring                        = false -> null
  - placement_partition_number        = 0 -> null
  - primary_network_interface_id      = "eni-08c3172a3d36b4f5f" -> null
  - private_dns                       = "ip-10-0-2-65.ap-south-1.compute.internal" -> null
```

```

- private_ip                = "10.0.2.65" -> null
- secondary_private_ips     = [] -> null
- security_groups           = [] -> null
- source_dest_check         = true -> null
- subnet_id                 = "subnet-03695136206a6d1c0" -> null
- tags                      = {
  - "Environment" = "dev"
  - "Name"        = "pavan-two-tier-db-server"
  - "Owner"       = "Pavan"
} -> null
- tags_all                  = {
  - "Environment" = "dev"
  - "Name"        = "pavan-two-tier-db-server"
  - "Owner"       = "Pavan"
} -> null
- tenancy                   = "default" -> null
- user_data_replace_on_change = false -> null
- vpc_security_group_ids    = [
  - "sg-0ef96a959e337c451",
] -> null
# (10 unchanged attributes hidden)

- capacity_reservation_specification {
  - capacity_reservation_preference = "open" -> null
}

- cpu_options {
  - core_count      = 1 -> null
  - threads_per_core = 2 -> null
  # (1 unchanged attribute hidden)
}

- credit_specification {
  - cpu_credits = "unlimited" -> null
}

- enclave_options {
  - enabled = false -> null
}

- maintenance_options {
  - auto_recovery = "default" -> null
}

- metadata_options {
  - http_endpoint = "enabled" -> null
}

```

```

- http_protocol_ipv6          = "disabled" -> null
- http_put_response_hop_limit = 2 -> null
- http_tokens                 = "required" -> null
- instance_metadata_tags      = "disabled" -> null
}

- private_dns_name_options {
  - enable_resource_name_dns_a_record    = false -> null
  - enable_resource_name_dns_aaaa_record = false -> null
  - hostname_type                       = "ip-name" -> null
}

- root_block_device {
  - delete_on_termination = true -> null
  - device_name           = "/dev/xvda" -> null
  - encrypted             = false -> null
  - iops                  = 3000 -> null
  - tags                  = {} -> null
  - tags_all              = {} -> null
  - throughput            = 125 -> null
  - volume_id             = "vol-0017b0496da84deec" -> null
  - volume_size           = 8 -> null
  - volume_type            = "gp3" -> null
  # (1 unchanged attribute hidden)
}
}

# aws_instance.web will be destroyed
- resource "aws_instance" "web" {
  - ami                      = "ami-03054015e26069645" -> null
  - arn                     = "arn:aws:ec2:ap-south-1:987078319571:instance/i-009a7068ece7510db" -> null
  - associate_public_ip_address = true -> null
  - availability_zone         = "ap-south-1a" -> null
  - cpu_core_count            = 1 -> null
  - cpu_threads_per_core      = 2 -> null
  - disable_api_stop          = false -> null
  - disable_api_termination   = false -> null
  - ebs_optimized             = false -> null
  - get_password_data         = false -> null
  - hibernation               = false -> null
  - id                       = "i-009a7068ece7510db" -> null
  - instance_initiated_shutdown_behavior = "stop" -> null
  - instance_state            = "running" -> null
  - instance_type             = "t3.micro" -> null
  - ipv6_address_count         = 0 -> null
  - ipv6_addresses            = [] -> null

```

```

- monitoring                = false -> null
- placement_partition_number = 0 -> null
- primary_network_interface_id = "eni-0cd3ed352c4badad4" -> null
- private_dns                = "ip-10-0-1-178.ap-south-1.compute.internal" -> null
- private_ip                 = "10.0.1.178" -> null
- public_dns                  = "ec2-13-206-201-23.ap-south-1.compute.amazonaws.com" -> null
- public_ip                   = "13.206.201.23" -> null
- secondary_private_ips       = [] -> null
- security_groups              = [] -> null
- source_dest_check            = true -> null
- subnet_id                    = "subnet-0e012946894175486" -> null
- tags                         = {
  - "Environment" = "dev"
  - "Name"         = "pavan-two-tier-web-server"
  - "Owner"        = "Pavan"
} -> null
- tags_all                    = {
  - "Environment" = "dev"
  - "Name"         = "pavan-two-tier-web-server"
  - "Owner"        = "Pavan"
} -> null
- tenancy                     = "default" -> null
- user_data                    = "2a6dcb73cb3ff7fe2af9f6bee13a158d650f8000" -> null
- user_data_replace_on_change = true -> null
- vpc_security_group_ids       = [
  - "sg-0cfff1c26ee7cbdfb9",
] -> null
# (8 unchanged attributes hidden)

- capacity_reservation_specification {
  - capacity_reservation_preference = "open" -> null
}

- cpu_options {
  - core_count      = 1 -> null
  - threads_per_core = 2 -> null
  # (1 unchanged attribute hidden)
}

- credit_specification {
  - cpu_credits = "unlimited" -> null
}

- enclave_options {
  - enabled = false -> null
}

```

```

- maintenance_options {
  - auto_recovery = "default" -> null
}

- metadata_options {
  - http_endpoint          = "enabled" -> null
  - http_protocol_ipv6     = "disabled" -> null
  - http_put_response_hop_limit = 2 -> null
  - http_tokens            = "required" -> null
  - instance_metadata_tags  = "disabled" -> null
}

- private_dns_name_options {
  - enable_resource_name_dns_a_record    = false -> null
  - enable_resource_name_dns_aaaa_record = false -> null
  - hostname_type                       = "ip-name" -> null
}

- root_block_device {
  - delete_on_termination = true -> null
  - device_name           = "/dev/xvda" -> null
  - encrypted             = false -> null
  - iops                  = 3000 -> null
  - tags                  = {} -> null
  - tags_all              = {} -> null
  - throughput            = 125 -> null
  - volume_id             = "vol-074fbddb2e3f05060" -> null
  - volume_size           = 8 -> null
  - volume_type           = "gp3" -> null
  # (1 unchanged attribute hidden)
}
}

# aws_internet_gateway.main will be destroyed
- resource "aws_internet_gateway" "main" {
  - arn      = "arn:aws:ec2:ap-south-1:987078319571:internet-gateway/igw-064471428d88acd6a" -> null
  - id      = "igw-064471428d88acd6a" -> null
  - owner_id = "987078319571" -> null
  - tags    = {
    - "Environment" = "dev"
    - "Name"        = "pavan-two-tier-igw"
    - "Owner"       = "Pavan"
  } -> null
  - tags_all = {
    - "Environment" = "dev"

```

```

- "Name"          = "pavan-two-tier-igw"
- "Owner"         = "Pavan"
} -> null
- vpc_id          = "vpc-0fc2b2a37e1ef0efc" -> null
}

# aws_route_table.public will be destroyed
- resource "aws_route_table" "public" {
-   arn            = "arn:aws:ec2:ap-south-1:987078319571:route-table/rtb-0d4800a7500ab5398" -> null
-   id             = "rtb-0d4800a7500ab5398" -> null
-   owner_id       = "987078319571" -> null
-   propagating_vgws = [] -> null
-   route          = [
-     {
-       cidr_block      = "0.0.0.0/0"
-       gateway_id      = "igw-064471428d88acd6a"
-       # (11 unchanged attributes hidden)
-     },
-   ] -> null
-   tags           = {
-     "Environment" = "dev"
-     "Name"        = "pavan-two-tier-public-rt"
-     "Owner"       = "Pavan"
-   } -> null
-   tags_all       = {
-     "Environment" = "dev"
-     "Name"        = "pavan-two-tier-public-rt"
-     "Owner"       = "Pavan"
-   } -> null
-   vpc_id         = "vpc-0fc2b2a37e1ef0efc" -> null
}

# aws_route_table_association.public will be destroyed
- resource "aws_route_table_association" "public" {
-   id             = "rtbassoc-08ec72c09adc1b2f4" -> null
-   route_table_id = "rtb-0d4800a7500ab5398" -> null
-   subnet_id      = "subnet-0e012946894175486" -> null
-   # (1 unchanged attribute hidden)
}

# aws_security_group.db will be destroyed
- resource "aws_security_group" "db" {
-   arn            = "arn:aws:ec2:ap-south-1:987078319571:security-group/sg-0ef96a959e337c451" -> null
-   description    = "Allow MySQL only from the web security group" -> null
-   egress         = [
-     {

```

```

        - cidr_blocks      = [
          - "0.0.0.0/0",
        ]
        - description      = "Allow outbound traffic"
        - from_port        = 0
        - ipv6_cidr_blocks = []
        - prefix_list_ids  = []
        - protocol         = "-1"
        - security_groups  = []
        - self             = false
        - to_port          = 0
      },
    ] -> null
  - id              = "sg-0ef96a959e337c451" -> null
  - ingress         = [
    - {
      - cidr_blocks      = []
      - description      = "MySQL from web server security group"
      - from_port        = 3306
      - ipv6_cidr_blocks = []
      - prefix_list_ids  = []
      - protocol         = "tcp"
      - security_groups  = [
        - "sg-0cfff1c26ee7cbdfb9",
      ]
      - self             = false
      - to_port          = 3306
    },
  ] -> null
  - name            = "pavan-two-tier-db-sg" -> null
  - owner_id        = "987078319571" -> null
  - revoke_rules_on_delete = false -> null
  - tags            = {
    - "Environment" = "dev"
    - "Name"        = "pavan-two-tier-db-sg"
    - "Owner"       = "Pavan"
  } -> null
  - tags_all        = {
    - "Environment" = "dev"
    - "Name"        = "pavan-two-tier-db-sg"
    - "Owner"       = "Pavan"
  } -> null
  - vpc_id          = "vpc-0fc2b2a37e1ef0efc" -> null
  # (1 unchanged attribute hidden)
}

```

aws_security_group.web will be destroyed

```
- resource "aws_security_group" "web" {
  - arn                = "arn:aws:ec2:ap-south-1:987078319571:security-group/sg-0cff1c26ee7cbdfb9" -> null
  - description        = "Allow HTTP and restricted SSH access" -> null
  - egress              = [
    - {
      - cidr_blocks      = [
        - "0.0.0.0/0",
      ]
      - description      = "Allow outbound traffic"
      - from_port         = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids  = []
      - protocol          = "-1"
      - security_groups  = []
      - self              = false
      - to_port           = 0
    },
  ] -> null
  - id                  = "sg-0cff1c26ee7cbdfb9" -> null
  - ingress              = [
    - {
      - cidr_blocks      = [
        - "0.0.0.0/0",
      ]
      - description      = "HTTP from the internet"
      - from_port         = 80
      - ipv6_cidr_blocks = []
      - prefix_list_ids  = []
      - protocol          = "tcp"
      - security_groups  = []
      - self              = false
      - to_port           = 80
    },
  ]
  - {
    - cidr_blocks      = [
      - "106.222.228.188/32",
    ]
    - description      = "SSH from my IP only"
    - from_port         = 22
    - ipv6_cidr_blocks = []
    - prefix_list_ids  = []
    - protocol          = "tcp"
    - security_groups  = []
    - self              = false
    - to_port           = 22
  }
```

```

    },
  ] -> null
- name                = "pavan-two-tier-web-sg" -> null
- owner_id            = "987078319571" -> null
- revoke_rules_on_delete = false -> null
- tags                = {
  - "Environment" = "dev"
  - "Name"        = "pavan-two-tier-web-sg"
  - "Owner"       = "Pavan"
} -> null
- tags_all            = {
  - "Environment" = "dev"
  - "Name"        = "pavan-two-tier-web-sg"
  - "Owner"       = "Pavan"
} -> null
- vpc_id              = "vpc-0fc2b2a37e1ef0efc" -> null
# (1 unchanged attribute hidden)
}

```

aws_subnet.private will be destroyed

```

- resource "aws_subnet" "private" {
  - arn                  = "arn:aws:ec2:ap-south-1:987078319571:subnet/subnet-03695136206a6d1c0" -> null
  - assign_ipv6_address_on_creation = false -> null
  - availability_zone      = "ap-south-1a" -> null
  - availability_zone_id   = "aps1-az1" -> null
  - cidr_block             = "10.0.2.0/24" -> null
  - enable_dns64           = false -> null
  - enable_lni_at_device_index = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                    = "subnet-03695136206a6d1c0" -> null
  - ipv6_native           = false -> null
  - map_customer_owned_ip_on_launch = false -> null
  - map_public_ip_on_launch = false -> null
  - owner_id              = "987078319571" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - tags                  = {
    - "Environment" = "dev"
    - "Name"        = "pavan-two-tier-private-subnet"
    - "Owner"       = "Pavan"
  } -> null
  - tags_all              = {
    - "Environment" = "dev"
    - "Name"        = "pavan-two-tier-private-subnet"
    - "Owner"       = "Pavan"
  } -> null
}

```

```

- vpc_id                                = "vpc-0fc2b2a37e1ef0efc" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.public will be destroyed
- resource "aws_subnet" "public" {
  - arn                                = "arn:aws:ec2:ap-south-1:987078319571:subnet/subnet-0e012946894175486" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "ap-south-1a" -> null
  - availability_zone_id                = "aps1-az1" -> null
  - cidr_block                         = "10.0.1.0/24" -> null
  - enable_dns64                       = false -> null
  - enable_lni_at_device_index         = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                 = "subnet-0e012946894175486" -> null
  - ipv6_native                        = false -> null
  - map_customer_owned_ip_on_launch    = false -> null
  - map_public_ip_on_launch            = true -> null
  - owner_id                           = "987078319571" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - tags                               = {
    - "Environment" = "dev"
    - "Name"         = "pavan-two-tier-public-subnet"
    - "Owner"        = "Pavan"
  } -> null
  - tags_all                               = {
    - "Environment" = "dev"
    - "Name"         = "pavan-two-tier-public-subnet"
    - "Owner"        = "Pavan"
  } -> null
  - vpc_id                                = "vpc-0fc2b2a37e1ef0efc" -> null
  # (4 unchanged attributes hidden)
}

# aws_vpc.main will be destroyed
- resource "aws_vpc" "main" {
  - arn                                = "arn:aws:ec2:ap-south-1:987078319571:vpc/vpc-0fc2b2a37e1ef0efc" -> null
  - assign_generated_ipv6_cidr_block    = false -> null
  - cidr_block                         = "10.0.0.0/16" -> null
  - default_network_acl_id              = "acl-06e0f838e36e72a84" -> null
  - default_route_table_id              = "rtb-099db34e641721011" -> null
  - default_security_group_id           = "sg-04ac0de2ce9827ef4" -> null
  - dhcp_options_id                     = "dopt-04a135d49bd3928a4" -> null
  - enable_dns_hostnames                = true -> null
  - enable_dns_support                  = true -> null

```

```

- enable_network_address_usage_metrics = false -> null
- id                                   = "vpc-0fc2b2a37e1ef0efc" -> null
- instance_tenancy                     = "default" -> null
- ipv6_netmask_length                  = 0 -> null
- main_route_table_id                  = "rtb-099db34e641721011" -> null
- owner_id                             = "987078319571" -> null
- tags                                 = {
  - "Environment" = "dev"
  - "Name"         = "pavan-two-tier-vpc"
  - "Owner"        = "Pavan"
} -> null
- tags_all                                     = {
  - "Environment" = "dev"
  - "Name"         = "pavan-two-tier-vpc"
  - "Owner"        = "Pavan"
} -> null
# (4 unchanged attributes hidden)
}

```

Plan: 0 to add, 0 to change, 10 to destroy.

Changes to Outputs:

```

- db_private_ip = "10.0.2.65" -> null
- vpc_id        = "vpc-0fc2b2a37e1ef0efc" -> null
- web_public_ip = "13.206.201.23" -> null
- web_public_url = "http://13.206.201.23" -> null

```

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```

aws_route_table_association.public: Destroying... [id=rtbassoc-08ec72c09adc1b2f4]
aws_instance.db: Destroying... [id=i-02e89085cd19517c6]
aws_instance.web: Destroying... [id=i-009a7068ece7510db]
aws_route_table_association.public: Destruction complete after 1s
aws_route_table.public: Destroying... [id=rtb-0d4800a7500ab5398]
aws_route_table.public: Destruction complete after 0s
aws_internet_gateway.main: Destroying... [id=igw-064471428d88acd6a]
aws_instance.web: Still destroying... [id=i-009a7068ece7510db, 00m10s elapsed]
aws_instance.db: Still destroying... [id=i-02e89085cd19517c6, 00m10s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-064471428d88acd6a, 00m10s elapsed]
aws_internet_gateway.main: Destruction complete after 18s
aws_instance.db: Still destroying... [id=i-02e89085cd19517c6, 00m20s elapsed]
aws_instance.web: Still destroying... [id=i-009a7068ece7510db, 00m20s elapsed]

```

```
aws_instance.web: Still destroying... [id=i-009a7068ece7510db, 00m30s elapsed]
aws_instance.db: Still destroying... [id=i-02e89085cd19517c6, 00m30s elapsed]
aws_instance.web: Destruction complete after 31s
aws_subnet.public: Destroying... [id=subnet-0e012946894175486]
aws_subnet.public: Destruction complete after 0s
aws_instance.db: Still destroying... [id=i-02e89085cd19517c6, 00m40s elapsed]
aws_instance.db: Destruction complete after 41s
aws_subnet.private: Destroying... [id=subnet-03695136206a6d1c0]
aws_security_group.db: Destroying... [id=sg-0ef96a959e337c451]
aws_subnet.private: Destruction complete after 0s
aws_security_group.db: Destruction complete after 0s
aws_security_group.web: Destroying... [id=sg-0cfff1c26ee7cbdfb9]
aws_security_group.web: Destruction complete after 1s
aws_vpc.main: Destroying... [id=vpc-0fc2b2a37elef0efc]
aws_vpc.main: Destruction complete after 1s
Releasing state lock. This may take a few moments...
```

Destroy complete! Resources: 10 destroyed.